FUNCTORIALITY OF THE SMALL ETALE SITE OF A FORMAL ALGEBRAIC SPACE

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file formal-spaces.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof span cites one same-chapter lemma and one cross-chapter lemma of Properties of Spaces, both delivered as navigation stubs; the statement span cites one definition of the chapter Sites, also delivered as a navigation stub. No bibliography citation occurs in the spans.

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Allowed standard prerequisites (editorial).

- The reduction of a formal algebraic space, which is an algebraic space, and its functoriality in the morphism f. Same chapter, Lemma lemma-reduction-formal-algebraic-space, invoked at proof lines 7441-7442. Produces the morphism of reductions that reduces the claim to the algebraic-space case.
- Functoriality of the small etale site for a morphism of algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-functoriality-etale-site, invoked at proof lines 7443-7445. Supplies the continuous functor, the morphism of sites, and the compatibility with compositions.
- The comparison between the etale site of a formal algebraic space and that of its reduction. Same chapter, equation (equation-etale-topos) and the surrounding construction of the small etale topos. Turns the morphism of sites into the morphism of topoi f_small of item (3).

Source reference. Original equation 0DEB: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 7413–7413. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original lemma 0AIN: sources/stacks-formal-spaces/pinned-original-source.tex, source lines 2833–2833. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Source reference. Original definition 03CC: sources/stacks-chow/tags.txt, source lines 4379–4379. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Date: stacks-project:formal-spaces.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 03G2: sources/stacks-chow/tags.txt, source lines 4513–4513. Pointer outside the counted span, invoked once for context; the load-bearing argument of the counted statement is the author's proof reproduced in full in the proof span.

Lemma 0.1. *Let S be a scheme. Let $f : X \to Y$ be a morphism of formal algebraic spaces over S.*

(1) *There is a continuous functor $Y_{spaces,étale} \to X_{spaces,étale}$ which induces a morphism of sites*
$$f_{spaces,étale} : X_{spaces,étale} \to Y_{spaces,étale}.$$
(2) *The rule $f \mapsto f_{spaces,étale}$ is compatible with compositions, in other words $(f \circ g)_{spaces,étale} = f_{spaces,étale} \circ g_{spaces,étale}$ (see Sites, Definition 03CC).*
(3) *The morphism of topoi associated to $f_{spaces,étale}$ induces, via (0DEB), a morphism of topoi $f_{small} : Sh(X_{étale}) \to Sh(Y_{étale})$ whose construction is compatible with compositions.*

Proof. The only point here is that f induces a morphism of reductions $X_{red} \to Y_{red}$ by Lemma 0AIN. Hence this lemma is immediate from the corresponding lemma for morphisms of algebraic spaces (Properties of Spaces, Lemma 03G2). □